

Samvardhini Sridharan

Postdoctoral Researcher

University of Washington School of Medicine, Genome Sciences

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Education

Ph.D. in Molecular and Cell Biology with a Designated Emphasis in Genomics and Computational Biology,

University of California – Berkeley

Aug 2020 – Dec 2025

Thesis: *Recurrent Structural Variation and Inversion Haplotype Diversity in Humans and Great Apes*

Thesis supervisor: Dr. Peter Sudmant

Committee members: Drs. Craig Miller, Rasmus Nielsen, Brenna Henn, Priya Moorjani

B.S. in Genetics and Genomics, University of California – Davis

Sep 2016 – Jun 2020

Minors: Statistics and Professional Writing

Research Interests

- Structural variation
- Genome Wide Association Studies (GWAS)
- Primate genomics
- Population genomics

Research and Industry Experience

Postdoctoral Researcher, University of Washington, Seattle

Mar 2026 – Present

Advisor: Dr. Saori Sakaue

- Developing a genome-wide inversion reference panel and imputation pipeline to enable structural variant association testing across biobank-scale cohorts, including the UK Biobank and *All of Us*.
- Constructing haplotype-resolved assembly alignment pipelines (minimap2, Snakemake) against the CHM13 reference to genotype inversion loci across HGSVC and HPRC cohorts.
- Collaborating with the Eichler lab on inversion callset validation and orthogonal confirmation of ambiguous haplotype calls.

Graduate Student Researcher, University of California, Berkeley

Aug 2020 – Dec 2025

Advisor: Dr. Peter Sudmant

- Explored structural variation, genetic diversity, comparative genomics, and evolution using computational tools and comparative genomics approaches with a primary focus on humans and non-human primates.
- Worked on large data-driven projects involving cutting-edge long-read sequencing data, collaboration with data consortiums (including HGSVC, HPRC, T2T), and utilization of genetic biobanks, with upwards of 5500 samples.

- Awarded a Leakey Foundation Fellowship to generate and analyze sequencing data in four new gorilla haplotypes to understand their implications in health and evolutionary origins.
- Analyzed and interpreted structural variation patterns across temporal and geographic scales using population genetic packages and original methods, successfully uncovering genetic signals in datasets that have been previously unexplored.
- Managed and utilized cluster computing environments for large-scale data analysis and computation, with a special focus on working with confidential and protected datasets.
- Automated data processing, analysis pipelines, and repetitive tasks using Snakemake to design reproducible and scalable computational workflows which are easily translatable to other genetic regions and species.

Genentech Biological Technologies Intern, Genentech, South San Francisco
Supervisor: Brian Murphy

Jun 2019 – Sep 2019

- Optimized data transformation and visualization processes by developing computational pipelines in R, Tableau, and PowerBI to streamline the electronic transformation of two decades worth of assay data, resulting in substantial data entry time savings, and reduced the risk of errors due to manual data entry.
- Created an app using R Shiny to automate paperwork and template filing, saving the team time, improving accuracy and reduced administrative overhead per research tech.
- Selected as part of a highly selective national 20-person cohort (gLinx) of undergraduate and graduate trainees to gain valuable insights on industry research and postgraduate research.

Canary Center at Stanford for Early Cancer Detection Intern, Stanford University, Palo Alto Jun 2018 – Sep 2018

Advisor: Dr. Ahmed El Kaffas

- Designed, fabricated, and prototyped a specialized biomedical device known as a “flow phantom” to investigate highly dynamic and variable ultrasound microbubbles, demonstrating promising results in enhancing medical imaging, drug delivery, and targeted therapies, while minimizing invasiveness and clinical research overheads.
- Used cost efficient and creative approaches to mimic tumor vasculature including using spun sugar (“cotton candy”) optimizing research and personnel spending.
- Participated and observed radiological data collection efforts, using tools in R, MATLAB, and Python to analyze ultrasonication tools and devices that would improve patient outcomes in clinical settings.
- Collected and analyzed primary literature in the nascent field of “radiogenomics,” which was awarded the 1st place Norma J. Lang Prize on Information Research.

Peer-Reviewed Publications

Sridharan, S., Lou, R.N., Ferguson, S., et al. (2026). Recurrent structural variation and recent turnover at the 17q21.31 locus in humans and great apes. *Nat Commun*, 1(2), 123–123. <https://doi.org/10.1038/s41467-026-73174-1>

Preprints and Manuscripts in Preparation

Sridharan, S., Ferguson, S., Sudmant, P.H. [Manuscript in preparation].

El Kaffas, A., Gong, R., **Sridharan, S.**, et al. (2020). Clinical Evaluation of Real-Time Optical-Tracking Navigation and Live Time-Intensity Curves to Provide Feedback During Blinded 4D Contrast-Enhanced Ultrasound Imaging. <https://doi.org/10.48550/arXiv.2011.00744>

Fellowships, Awards, and Honors

Trainee Travel Award, Long Read Sequencing Workshop	2026
Leakey Foundation Research Grant for Advanced Doctoral Students	2023
National Science Foundation Graduate Research Fellowship Program (NSF-GRFP) Honorable Mention	2022
UC Davis Outstanding Performance Citation, Departmental Citation, and Distinguished Scholar Award	2020
1st Place Norma J. Lang Prize for Undergraduate Information Research	2019
Aggie Alumni Advancing STEM Award	2019
Genentech-USBTD Outstanding Student in Molecular and Cell Biology	2018

Selected Conference Presentations

<i>Oral Presentation</i> , Long Read Sequencing Workshop, Jackson Laboratories, Farmington, CT	May 2026
<i>Poster Presentation</i> , Long Read Sequencing Workshop, Jackson Laboratories, Farmington, CT	May 2026
<i>Oral Presentation</i> , Bay Area Population Genetics, Stanford University	Dec 2025
<i>Oral Presentation</i> , Center for Computational Biology Retreat, Los Altos, CA	Oct 2025
<i>Oral Presentation</i> , Genetics, Genomics, Evolution, and Development Retreat, Berkeley, CA	Oct 2025
<i>Poster Presentation</i> , Biology of Genomes, Cold Spring Harbor Laboratory, New York	May 2025
<i>Poster Presentation</i> , Bay Area Population Genetics, University of California – Berkeley	Nov 2024
<i>Oral Presentation</i> , Society of Molecular and Cell Biology (SMBE), Puerto Vallarta, Mexico	Jul 2024
<i>Poster Presentation</i> , American Society of Human Genetics (ASHG), Washington D.C	Nov 2023
<i>Poster Presentation</i> , Bay Area Population Genetics, University of California – Davis	May 2023
<i>Poster Presentation</i> , Biology of Genomes, Cold Spring Harbor Laboratory, New York	May 2023
<i>Poster Presentation</i> , Computing Research Association Widening Participation (CRA-WP), New Orleans, LA	Apr 2022

Teaching Experience

Graduate Student Instructor, MCELLBI 104 — Genetics, Genomics and Cell Biology, University of California, Berkeley Fall 2022, Spring 2025, Fall 2025

- Facilitated two discussion sections each week (~25 students per section), utilizing diverse teaching methods including chalk talks, PowerPoint presentations, and interactive poll questions.
- Developed and prepared comprehensive discussion section materials, including worksheets, chalkboard activities, and presentations, designed to cater to individual, pair, and small group work.
- Created and reviewed exam questions to ensure alignment with section topics and learning objectives.
- Held twice-weekly office hours to address student inquiries, provide academic mentorship, and offer guidance on research, graduate school, and professional career paths.
- Collaborated with fellow GSIs and instructors to review lecture content, clarify challenging concepts, and brainstorm effective practice problems for upcoming sections.

Student Advising

Alicia (Ally) Aggarwal, Summer Undergraduate Research Volunteer, Cornell University Jun 2026 – Present

Emily Liao, Undergraduate Research Shadowing Program, UC Berkeley Oct 2024 – Dec 2024

Community Engagement

Blog Editor-in-Chief, Berkeley Science Nov 2021 – Dec 2023

Writer, and Editor, Berkeley Science Review Jul 2020 – Sep 2024

- Developed and executed a comprehensive content strategy for the Berkeley Science Review blog and print magazine by aligning topics and themes with the publication's mission and audience, which includes over 2000 readers online and in print.
- Provided editorial direction to a team of 10+ editors ensuring the publication maintained a high standard of quality, thorough research, and reader engagement to refine and communicate complex scientific concepts to a broader non-scientific audience.
- Contributed to over 10 diverse articles to Berkeley Science Review print magazine and blog with topics ranging from short, impactful pieces (50 words) to comprehensive feature articles (2000 words) on topics in machine learning, evolution, and human health. A portfolio to these pieces can be found here: <https://www.berkeleysciencereview.com/author/samvardhini-sridharan/>

Python Boot Camp Co-Director, Center for Computational Biology, UC Berkeley Jan 2024 – Jun 2025

Python Boot Camp Lecturer and Teaching Assistant Jan 2022 – Jun 2022

- Developed and refined teaching materials on Pandas and NumPy, designing more innovative and scaffolded examples to meet the evolving needs of learners, aiding with positive learning outcomes.
- Delivered engaging lectures to a cohort of 200 students, adeptly transitioning between in-person, virtual, and hybrid instruction, and tailored hands-on exercises to accommodate a diverse student body with varying bioinformatics skills.

Service

Center for Theoretical and Evolutionary Genomics (CTEG) Organizing Committee	2023 – 2025
Center for Computational Biology Seminar Series Organizing Committee	2022 – 2023